

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2

9 June 2025

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Mrs M Trollope

Moderator

Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 12 questions, 12 pages and a formula sheet. You may detach the formula sheet for your use.
Please ensure that your question paper is complete.
2. Answer ALL the questions in the answer books provided. Additional space is provided at the end of the answer book.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.

QUESTION 1

Khanyiso recorded the amount of data (in MB) he used during June. The information for the first 15 days is shown in the table below.

17	69	10	58	34	18	13	26	3	12	24	16	15	40	20
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- 1.1 For the data set above, calculate the:
- 1.1.1 Standard deviation. (1)
- 1.1.2 Mean. (1)
- 1.2 Determine the number of days which the amount of data used was greater than one standard deviation above the mean. (2)
- 1.3 Calculate the amount of data that Khanyiso can use for the rest of the month, if he wants the overall mean for June to be 20MB. (2)
- [6]**

QUESTION 2

The attendance of Grade 12 learners at Sunshine High School is recorded. The information in the table below represents the number of days those learners were absent during the year.

Number of days absent	Number of learners
$0 \leq x < 5$	30
$5 \leq x < 10$	41
$10 \leq x < 15$	94
$15 \leq x < 20$	47
$20 \leq x < 25$	11
$25 \leq x < 30$	7

- 2.1 Write down the modal class. (1)
- 2.2 How many learners were absent from school for less than 15 days? (1)
- 2.3 How many Grade 12 learners are at the school? (1)
- 2.4 Draw a cumulative frequency graph (ogive) to represent the data above on the grid provided in your answer book. (4)
- 2.5 Use the cumulative frequency graph to determine the median number of days the Grade 12 learners were absent. (2)

[9]

QUESTION 3

Mr Furlonger and Mr Grace recorded the wind speed and temperatures during the school's last 10 hockey games. The recorded data is shown in the table below.

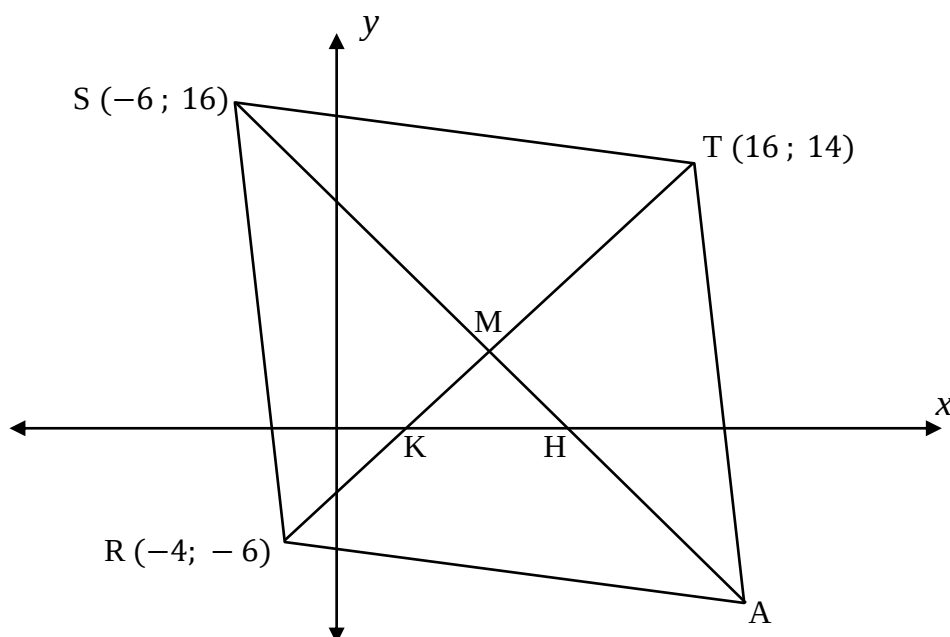
Wind Speed in km/h (x)	13	11	25	15	2	6	20	17	24	22
Temperature in °C (y)	19	19	26	24	16	22	28	26	22	20

- 3.1 Determine the equation of the least squares regression line for the data. (2)
- 3.2 Predict the temperature at the next hockey game if the wind speed during the game is 9 km per hour. (2)
- 3.3 Write down the correlation coefficient and comment on the correlation between wind speed and temperature. (2)

[6]

QUESTION 4

In the diagram below, $S(-6; 16)$, $T(16; 14)$, A and $R(-4; -6)$ are vertices of parallelogram STAR. The diagonals SA and RT intersect at M and cut the x -axis at H and K respectively.



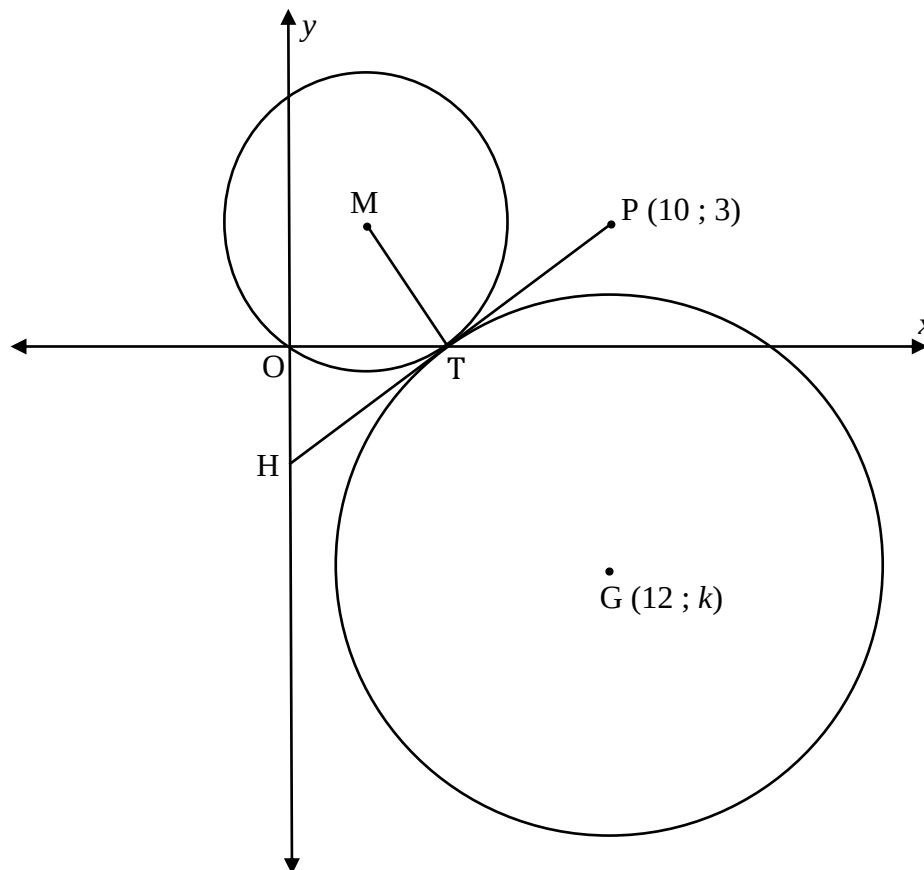
- 4.1 Calculate the coordinates of M. (2)
- 4.2 Calculate the gradient of RT. (2)
- 4.3 Determine the coordinates of A. (3)
- 4.4 Prove that STAR is a rhombus. (3)
- 4.5 Calculate the magnitude of $\widehat{MHK}$. (3)
- 4.6 The rhombus is reflected about the y -axis. N is the image of M after the reflection. Calculate the length of MN. (2)

[15]

QUESTION 5

In the diagram below is a circle centred at M with the equation $(x-3)^2 + (y-4)^2 = 25$.

This circle cuts the x -axis at the origin and T. HP is a tangent drawn from P (10 ; 3) to touch both circles at T and the y -axis at H. G (12 ; k) is the centre of the larger circle.



- 5.1 Write down the coordinates of M. (2)
- 5.2 Calculate the coordinates of T. (4)
- 5.3 Calculate the gradient of MT. (2)
- 5.4 Determine the equation of the tangent HP in the form $y = mx + c$. (4)
- 5.5 Another tangent to the circle with centre M is drawn from H to touch the circle at Q (not shown). Calculate the length of HQ. (4)
- 5.6 The circle with centre G touches the circle with centre M externally at T. Given that $MG = 15$ units, determine the value of k . (4)
- 5.7 Hence, write down the equation of the circle with centre G. (2)

[22]

QUESTION 6

6.1 If $\tan \theta = -\frac{5}{3}$ where $90^\circ < \theta < 270^\circ$, without the use of a calculator, and with the aid of a diagram, determine the value of $\cos 2\theta$. (6)

6.2 If $\sin 19^\circ = k$, determine the value of $\sin 38^\circ$ in terms of k . (3)

6.3 Simplify each of the following without the use of a calculator:

6.3.1
$$\frac{\sin(180^\circ - \theta) \cdot \tan(720^\circ + \theta) \cdot \cos(-\theta)}{-\cos(90^\circ + \theta) \cdot \tan(\theta - 180^\circ)}$$
 (6)

6.3.2
$$\frac{\cos 34^\circ \cos 10^\circ + \sin 34^\circ \sin 10^\circ}{\sin 12^\circ \cos 12^\circ}$$
 (4)

6.4 Prove the trigonometric identity below:

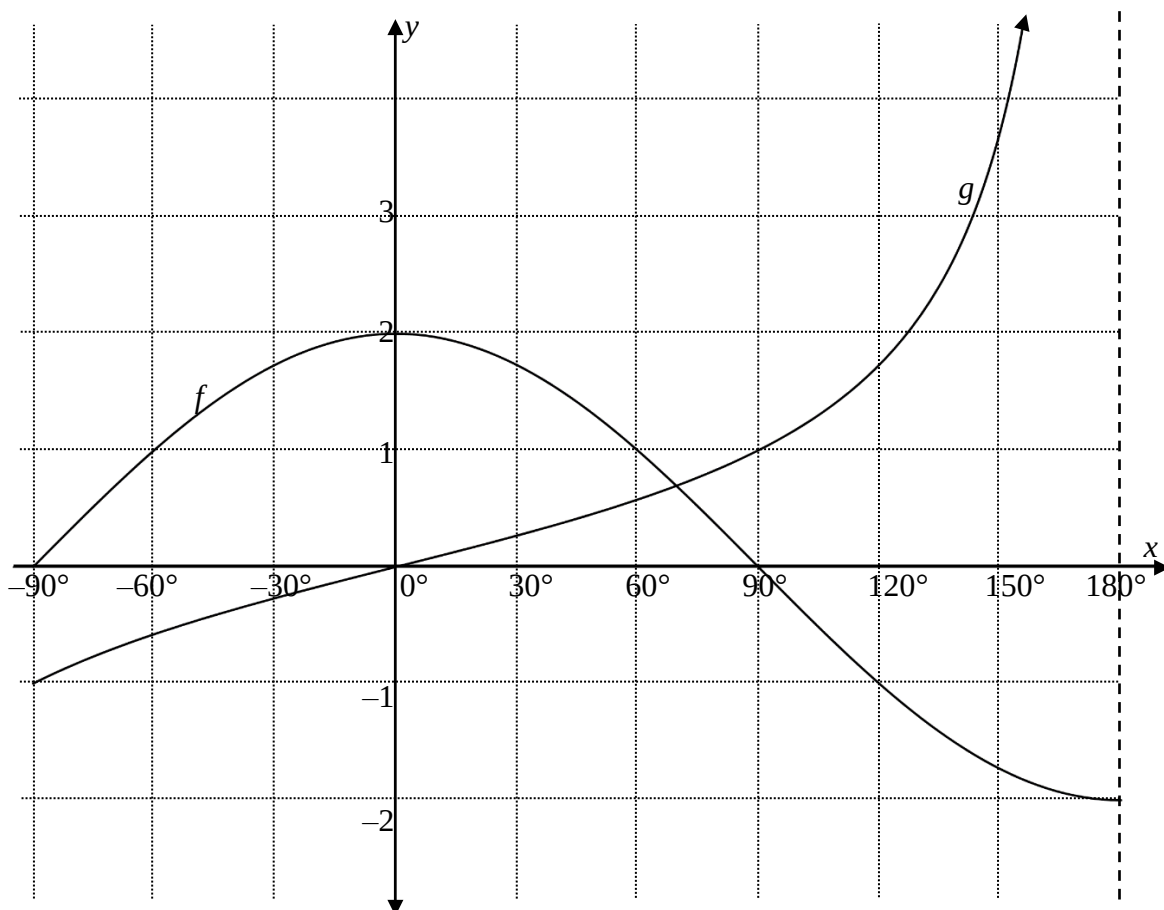
$$\frac{\sin 4x \cos 2x - 2 \cos 4x \sin x \cos x}{\tan 2x} = \cos^2 x - \sin^2 x. \quad (5)$$

6.5 Solve for x if $\sin 3x = -5 \cos 3x$, $x \in [0^\circ; 180^\circ]$. (6)

[30]

QUESTION 7

In the diagram below, the graphs of $f(x) = 2\cos x$ and $g(x) = \tan bx$ are drawn for the interval $x \in [-90^\circ; 180^\circ]$.



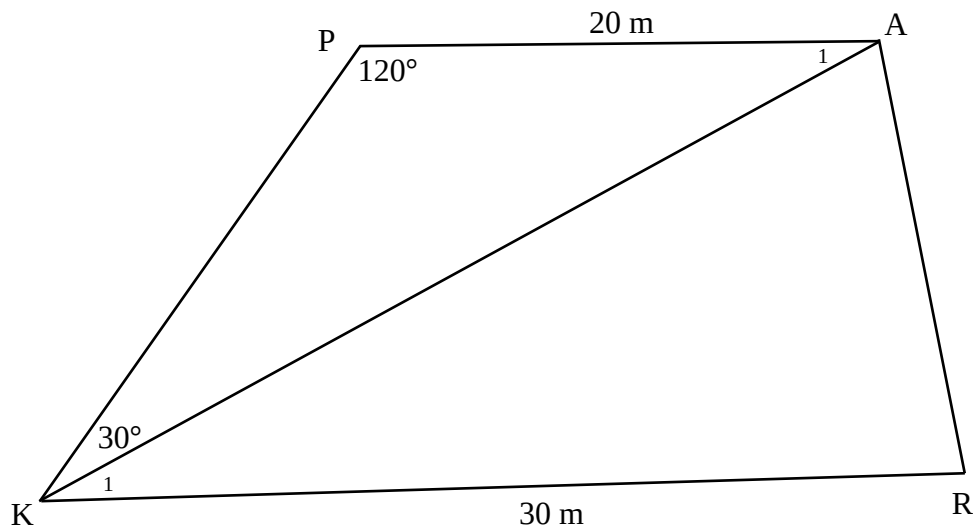
Use the graphs to answer the following questions:

- 7.1 Sketch the graph $m(x) = \sin 2x + 3$ on the set of axes provided, for the interval $x \in [-90^\circ; 180^\circ]$. (3)
- 7.2 Write down the value of b . (1)
- 7.3 Write down the range of g for the interval $x \in [-90^\circ; 180^\circ]$. (1)
- 7.4 Write down the period of g . (1)
- 7.5 The graph of h is obtained by shifting f 30° to the right and down by 4 units. Write down the equation of h . (2)
- 7.6 Use the graphs to determine for which values of $x \in [-90^\circ; 180^\circ]$ is $2\sin x \cos x > 0$. (2)

[10]

QUESTION 8

In the diagram below, PARK is a trapezium where $PA \parallel KR$. $PA = 20$ metres, $KR = 30$ metres, $\angle KPA = 120^\circ$ and $\angle PKA = 30^\circ$.



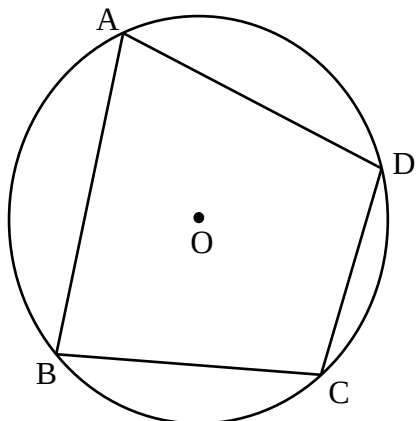
- 8.1 Find the length of KA, to the nearest metre. (3)
- 8.2 Find the length of AR, to the nearest metre. (4)
- 8.3 Find the area of PARK. (3)

[10]

Provide reasons for your statements in QUESTIONS 9, 10, 11 and 12.

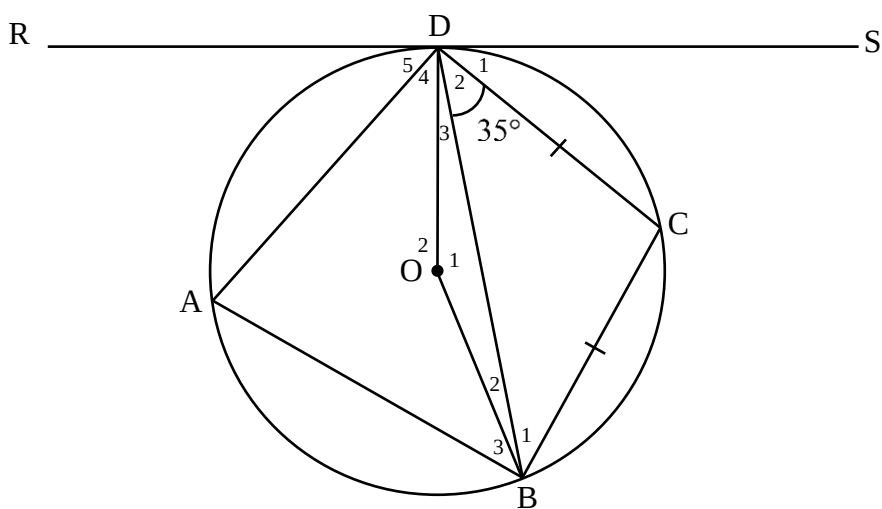
QUESTION 9

9.1 In the diagram below, ABCD is a cyclic quadrilateral and the circle has a centre O.



Prove the theorem which states that $\hat{A} + \hat{C} = 180^\circ$. (5)

9.2 In the diagram below, RDS is a tangent to the circle with centre O at D. $BC = DC$ and $\hat{CDB} = 35^\circ$.



Calculate with reasons the size of:

9.2.1 $\hat{D}_1$ (3)

9.2.2 $\hat{C}$ (2)

9.2.3 $\hat{A}$ (2)

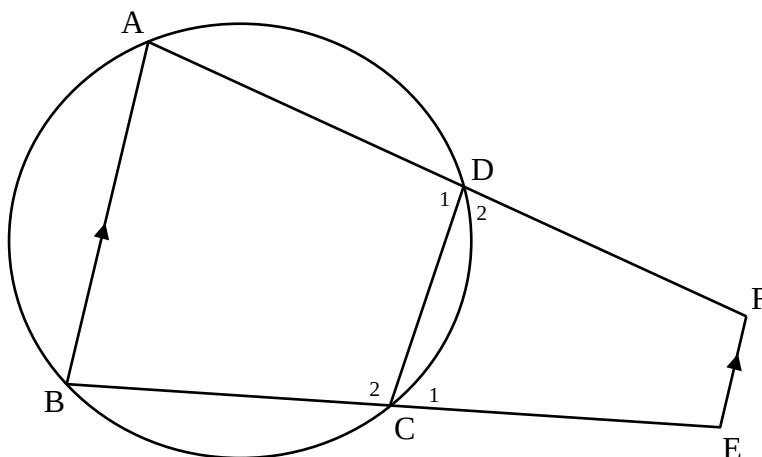
9.2.4 $\hat{O}_1$ (2)

9.2.5 $\hat{ODR}$ (2)

[16]

QUESTION 10

In the diagram below, ABCD is a cyclic quadrilateral. Chords AD and BC are produced to F and E, respectively. F and E are joined such that $EF \parallel AB$.

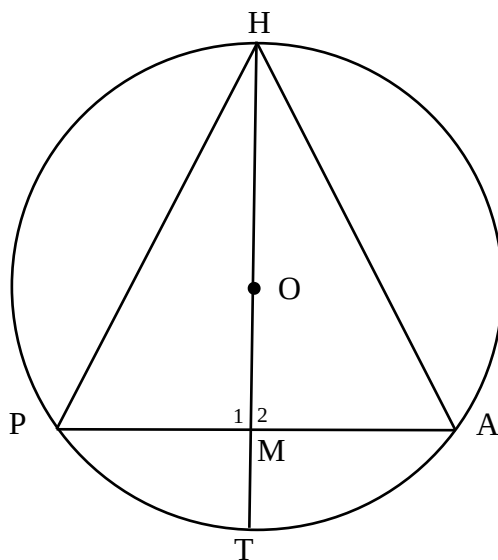


Prove that CEFD is a cyclic quadrilateral.

[4]

QUESTION 11

O is the centre of the circle. HM bisects PA. $PM = 12$ units, $MT = 6$ units, $HP = 12\sqrt{5}$ units and $OM = x$.



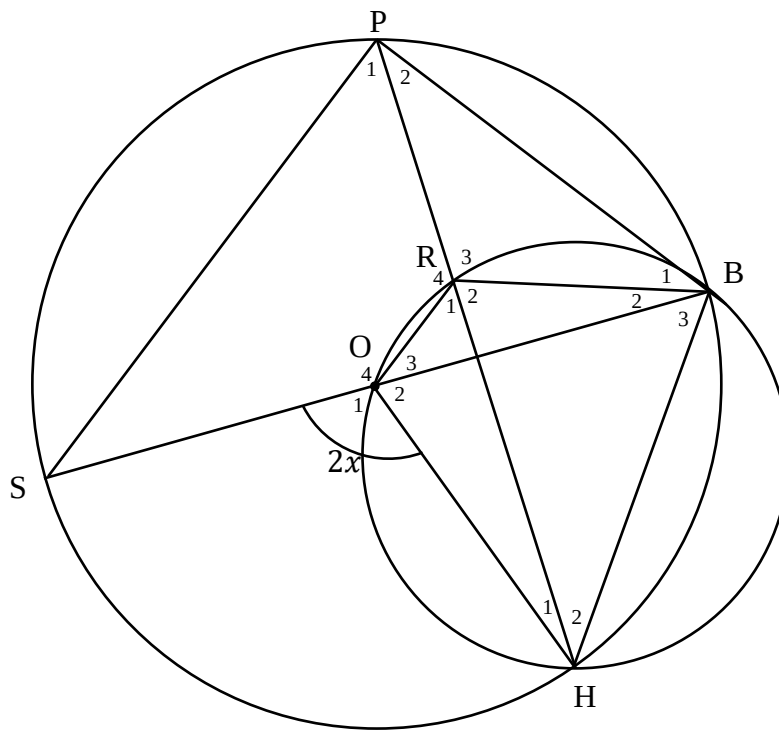
11.1 Express HM in terms of x . (2)

11.2 Calculate the area of the circle. (7)

[9]

QUESTION 12

In the sketch below, O is the centre of the larger circle. O lies on the circumference of the smaller circle. $\hat{O}_1 = 2x$.



12.1 State four angles equal to x . (4)

12.2 Determine the size of $\hat{R}_2$ in terms of x . (3)

12.3 Prove that $RP = RB$. (6)

[13]